Center for Data Analytics and Modeling

Chuka University

Machine Learning Assignment 1

Logistic Regression for CVD Prediction

Course: Machine Learning Methods
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Due Date: 1st April 2026

Learning Objectives

- Import and explore medical datasets in R
- Preprocess data for machine learning (cleaning, scaling, encoding)
- Implement train-test splitting and cross-validation
- Fit Logistic Regression models using `glm()` and `caret`
- Evaluate model performance using appropriate metrics
- Interpret model coefficients and odds ratios

1 General Instructions

- This assignment must be completed individually.
- Submit your work as a **PDF report** and the **R script** (.R or .Rmd).
- All code must be well-commented and reproducible.
- Use the provided dataset: `medical_data.csv`.
- Plots and tables must be clearly labeled.
- Late submissions will incur a penalty of 10% per day.

Important Note

The target variable is **CVD** (Cardiovascular Disease), coded as binary (0 = No, 1 = Yes). Ensure you convert this to a factor before modeling.

2 Dataset Description

The dataset `medical_data.csv` contains health indicators for 2,000 patients. The goal is to predict the presence of Cardiovascular Disease (CVD).

Variable	**Type**	**Description**
PatientID	Integer	Unique identifier
Age	Numeric	Age in years (18-80)
Gender	Factor	Male / Female
BMI	Numeric	Body Mass Index
SBP	Numeric	Systolic Blood Pressure
DBP	Numeric	Diastolic Blood Pressure
Cholesterol	Numeric	Total Cholesterol (mg/dL)
Glucose	Numeric	Fasting Glucose (mg/dL)
Smoking	Factor	Never / Former / Current / Occasional
Alcohol	Factor	None / Moderate / Heavy
Exercise	Factor	Sedentary / Light / Moderate / Vigorous
FamilyHistory	Factor	No / Yes
Medication	Factor	No / Yes
SleepHours	Numeric	Hours per night
StressLevel	Numeric	Scale 1-10
DietQuality	Numeric	Scale 1-5
CVD	**Binary**	**Target (0 = No, 1 = Yes)**

Table 1: Variable Description

3 Tasks

3.1 Task 1: Data Import and Exploration (10 Marks)

1. Load the necessary libraries (e.g., `tidyverse`, `caret`, `pROC`).
2. Import the dataset `medical_data.csv`.
3. Display the structure of the data (`str()`) and summary statistics (`summary()`).
4. Check for missing values and report the count per variable.
5. Visualize the distribution of the target variable **CVD** using a bar plot.

Code Hint

```
library(tidyverse)
library(caret)
data <- read_csv("medical_data.csv")
# Add your exploration code here
```

3.2 Task 2: Data Preprocessing (15 Marks)

1. Handle missing values (justify your method: removal or imputation).
2. Convert categorical variables (`Gender`, `Smoking`, `Alcohol`, `Exercise`, `FamilyHistory`, `Medication`) to factors.
3. Ensure the target variable `CVD` is a factor with levels `"No"` (0) and `"Yes"` (1).
4. Scale numeric predictors (mean = 0, sd = 1) using `preProcess()` from `caret`. **Note:** Fit the scaling parameters on the training data only.

3.3 Task 3: Data Partitioning (10 Marks)

1. Set the seed to `2026` for reproducibility.
2. Split the data into 70% training and 30% testing using stratified sampling (to preserve class balance).
3. Report the class distribution in both training and testing sets.

Code Hint

```
set.seed(2026)
train_index <- createDataPartition(data$CVD, p = 0.7, list =
    FALSE)
train_data <- data[train_index, ]
test_data <- data[-train_index, ]
```

3.4 Task 4: Model Training with trainControl (20 Marks)

1. Define a `trainControl` scheme with 10-fold cross-validation.
2. Train a Logistic Regression model using `caret::train()` with method `"glm"`.
3. Alternatively, fit a baseline model using base R `glm()` function.
4. Extract and report the model coefficients.

Code Hint

```
ctrl <- trainControl(method = "cv", number = 10,
                     classProbs = TRUE,
                     summaryFunction = twoClassSummary)
model <- train(CVD ~ ., data = train_data,
               method = "glm",
               trControl = ctrl,
               family = binomial)
```

3.5 Task 5: Model Evaluation (25 Marks)

1. Predict the class and probabilities on the **test set**.
2. Generate a Confusion Matrix and report:
   - Accuracy
   - Sensitivity (Recall)
   - Specificity
   - Kappa
3. Plot the ROC Curve and calculate the AUC (Area Under Curve).
4. Interpret the AUC value (e.g., poor, fair, good, excellent).

3.6 Task 6: Interpretation (20 Marks)

1. Select three significant predictors from your model.
2. Calculate the Odds Ratios (OR) for these predictors ($OR = e^{\beta}$).
3. Interpret the Odds Ratios in the context of CVD risk (e.g., "For every 1-unit increase in BMI, the odds of CVD increase by X%").
4. Discuss any limitations of your model (e.g., assumptions, potential biases).

4 Submission Guidelines

- **File Format:** Submit a single ZIP file named `ML_Assignment1_YourName.zip`.
- **Contents:**
  1. Report (PDF) containing answers, plots, and interpretations.
  2. R Script (.R) or R Markdown (.Rmd) with all code.
  3. The dataset (if modified).
- **Naming Convention:** `YourName_StudentID_ML1.pdf`
- **Submission Portal:** Upload to the CDAM Learning Management System.

5 Grading Rubric

Component	Marks	Criteria
Data Import & Exploration	10	Correct loading, summary, visualization
Preprocessing	15	Proper handling of NA, factors, scaling
Data Partitioning	10	Correct split, stratification, seed set
Model Training	20	Correct use of trainControl, glm/caret
Model Evaluation	25	Confusion matrix, ROC, AUC, metrics
Interpretation	20	Odds ratios, context, limitations
Total	**100**	

Table 2: Assignment Grading Rubric